LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Aerodynamic loading distribution, span wise and chord wise distribution, effects of sweep and distribution.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Aerodynamic loading distribution, span wise and chord wise distribution, effects of sweep and distribution.
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of wing structural elements and load paths using models; Instructor supervision required for model handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aircraft wing?
(b) Define the term 'aerodynamic loading'.
(c) List the main structural elements of a wing.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of aerodynamic loading • Span wise load distribution • Chord wise load distribution • Effect of wing sweep on loads	L & Ds	PPT, IB	15 min
(b)	• Structural response to span loads • Structural response to chord loads • Sweep effects on bending moments • Sweep effects on torsional loads	L & Ds	PPT, PDF	20 min
(c)	• Combined loading scenarios • Load factor implications • Design considerations for sweep • Summary of distribution effects	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how wing sweep affects the chord wise load distribution.
(b) Describe the relationship between span wise loading and structural elements.
(c) Differentiate between bending and torsional loads caused by sweep.

12. Home Assignment and Follow Up	Prepare a brief note on how the loading distribution principles discussed in this lesson apply to the miscellaneous loading cases covered in LP 08.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Miscellaneous loading cases, longitudinal cases, fuselage pressurization loads, flap and airbrake loads, spinning and power plants load, fuel systems, freight loads, ground handling guided missile launcher cases.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min